



HIGH CLOUD AIRLINES

A comprehensive data-driven review of our performance, key performance indicators (KPIs), and actionable insights leveraging modern BI tools.

Meet the **Analytics Team**

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Project Objective

The main goal of the High Cloud Airline Analytics project is to analyze airline data and generate meaningful insights that support better business decisions and improve overall airline performance.



Analyze operational and customer data to identify opportunities for increased efficiency and profitability.



Develop interactive dashboards for real-time performance tracking and reporting.



Identify top-performing routes, carriers, and emerging travel patterns and passenger trends.



Monitor key airline KPIs including Load Factor, Revenue, Flights, and Passenger Trends to enable data-driven decision-making.

Technology Stack



Power BI

Utilized as the primary visualization engine for executive summaries and high-level KPI tracking across the organization.



Tableau

Deployed for deep-dive exploratory data analysis, allowing regional managers to interactively slice and dice data.



Excel

Maintained for ad-hoc financial modeling, complex pivot table reporting, and legacy system data reconciliations.



SQL

The backbone of our data pipeline, used for complex querying, data cleaning, and transforming raw data into analytics-ready models.

1st KPI: Load Factor Percentage

Year	Load Factor %
[-] 2008	0.75
[+] Q1	0.70
[+] Q2	0.77
[+] Q3	0.78
[+] Q4	0.76
[+] 2009	0.75
[+] 2010	0.77
[+] 2011	0.78
[+] 2012	0.77
[+] 2013	0.78
Total	0.77

Key Performance Drivers



Consistent Growth: We can observe a general upward trend and stabilization in the Load Factor over time.



Early Dip: The years 2008 and 2009 were the lowest performing years in this dataset, both sitting at 0.75.



Peak Performance: The metric strengthened in the following years, peaking at 0.78 in both 2011 and 2013.



Stability: From 2010 onwards, the Load Factor remained highly consistent, fluctuating only slightly between 0.77 and 0.78.

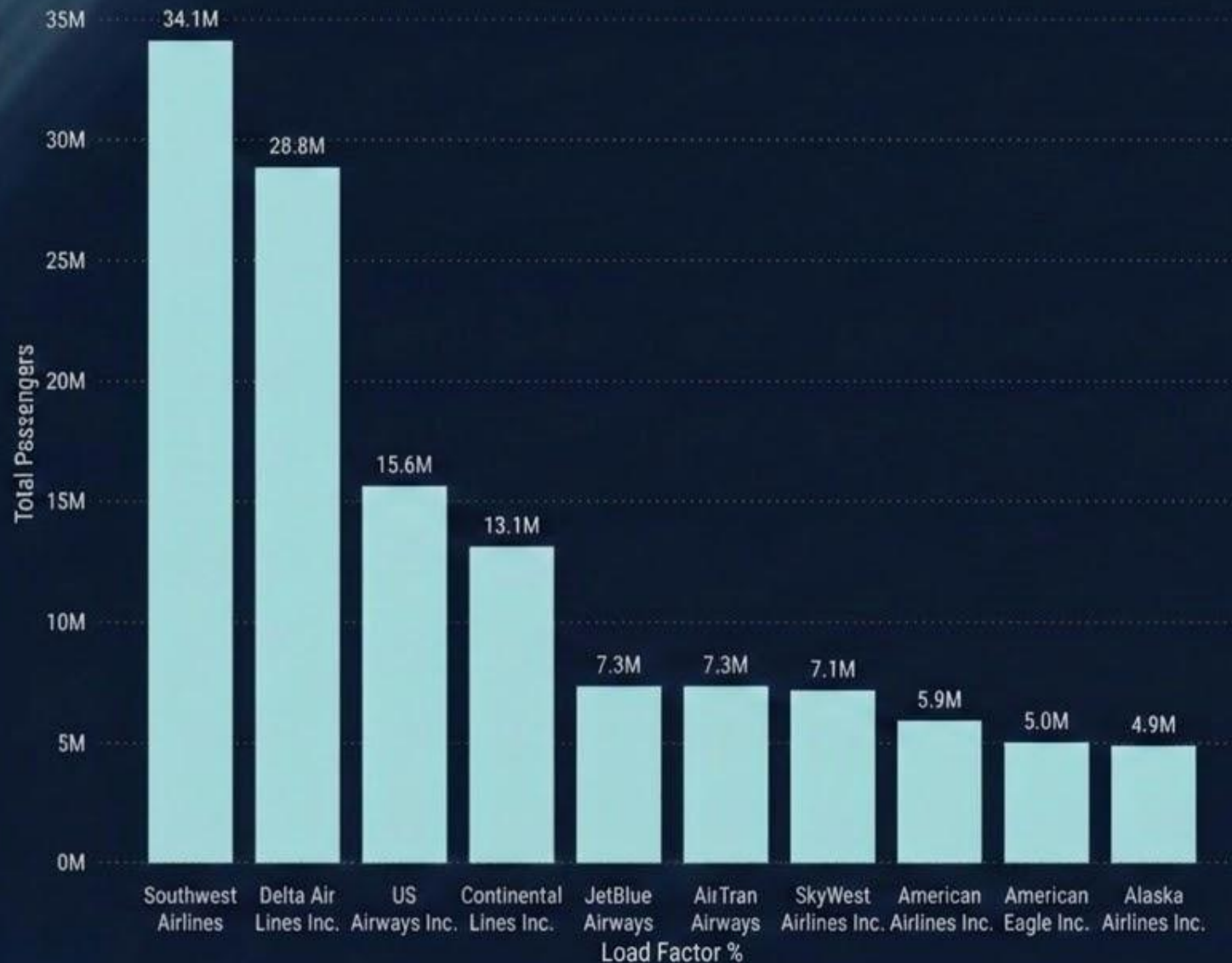
Load Factor % by Carrier Name



Strategic Load Factor Insights

- Load Factor Percentage represents the ratio of occupied seats to total available seats, indicating how efficiently the airline utilizes its seating capacity.
- Helps measure operational efficiency and flight occupancy performance.
- Identifies high-performing carriers with maximum seat utilization.
- Supports route optimization and capacity planning decisions.
- Assists in improving profitability by reducing unused seat inventory.
- Provides yearly and carrier-wise insights into passenger demand trends.

Total 10 Passengers by Carrier Name



Strategic Passenger Volume Insights



1. Market Demand & Presence

Represents total passengers transported by each airline, providing a clear measure of market demand and presence.



2. Operational Scale & Preference

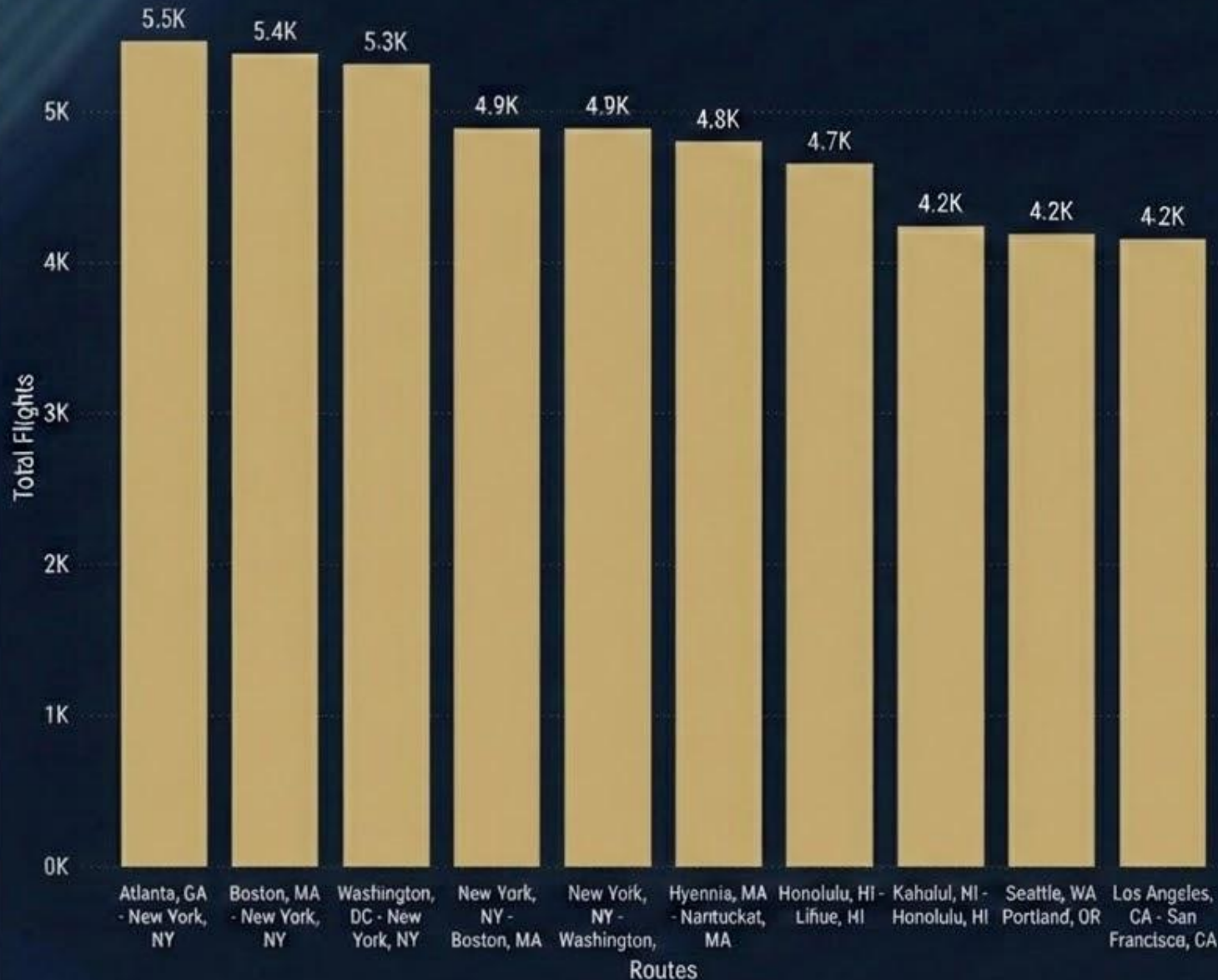
Airlines with higher volumes generally indicate stronger customer preference, wider networks, and greater operational scale.



3. Data-Driven Forecasting

Analyzing these traffic patterns assists in identifying leading carriers, forecasting trends, and strategic resource allocation for fleet and pricing.

Total 10 Flights by Routes



High-Frequency Route Optimization Strategy



1. Demand & Capacity Alignment

Analyze specific demand periods on top routes (e.g., ATL-NYC) to optimize aircraft size and flight frequency for maximized yield.



2. Connectivity Analysis

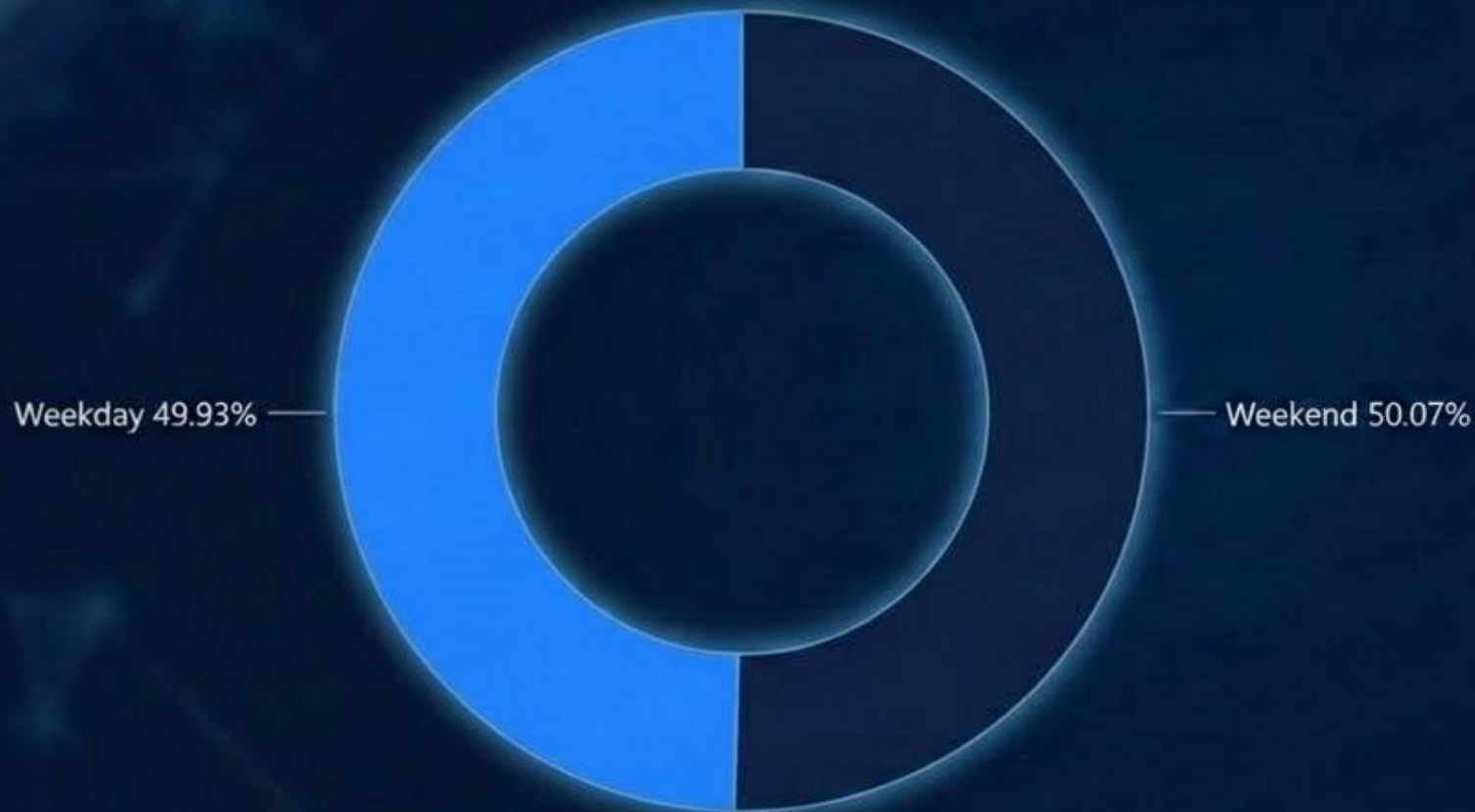
Examine connecting traffic flow at major hubs (e.g., Atlanta, Washington D.C.) to capture high-value connecting passengers.



3. Market Expansion

Utilize the unified SQL data warehouse (Ref. Slide 4, Point 4) to evaluate potential new high-frequency routes based on regional demand data.

Load Factor % by Day Type



Day Type Performance Analysis & Strategy



1. Balanced Flow: Over nearly 50/50 split indicating consistent demand and stable utilization of capacity.



2. Optimized Scheduling: Apply day-type data to improve fleet allocation and aircraft rotation schedules.



3. Segmented Promotions: Design targeted offers (weekday business vs. weekend leisure) for maximized yield.



4. Demand Forecasting: Leverage performance patterns to predict seasonal fluctuations and future passenger occupancy.

Total Flights by Distance Interval



KPIs & Key Insights

-  **SHORT-HAUL DOMINANCE:** Network structure is heavily defined by high-frequency shorter distance intervals, maximizing domestic connectivity.
-  **PREMIUM OPTIMIZATION:** Low volumes in long-distance segments signify key opportunities for premium route enhancement and targeted strategic expansion.
-  **RESOURCE EFFICIENCY ANALYSIS:** This metric enables critical evaluation of fleet deployment, fuel efficiency, and route profitability across all haul segments.



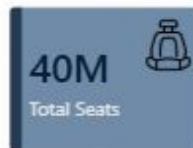
High Cloud Airlines

overview of airline operations and performance



Distance Interval

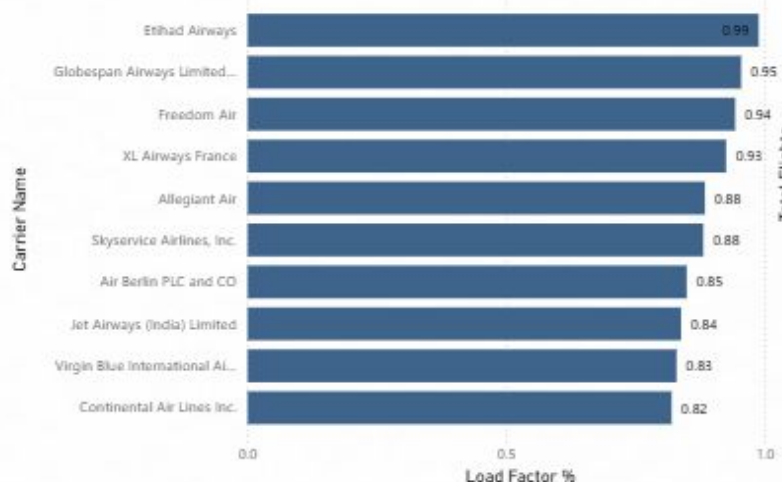
- ☐ 1000-1499 Miles
- ☐ 1500-1999 Miles
- ☐ 2000-2499 Miles
- ☐ 2500-2999 Miles
- ☐ 3000-3499 Miles
- ☐ 3500-3999 Miles
- ☐ 4000-4499 Miles
- ☐ 4500-4999 Miles
- ☐ 5000-5499 Miles
- ☐ 500-999 Miles
- ☐ 5500-5999 Miles
- ☐ 6000-6499 Miles
- ☐ 6500-6999 Miles
- ☐ 7000-7499 Miles
- ☐ 7500-7999 Miles
- ☐ 8000-8499 Miles
- ☐ 8500-8999 Miles
- ☐ Less Than 500 Miles



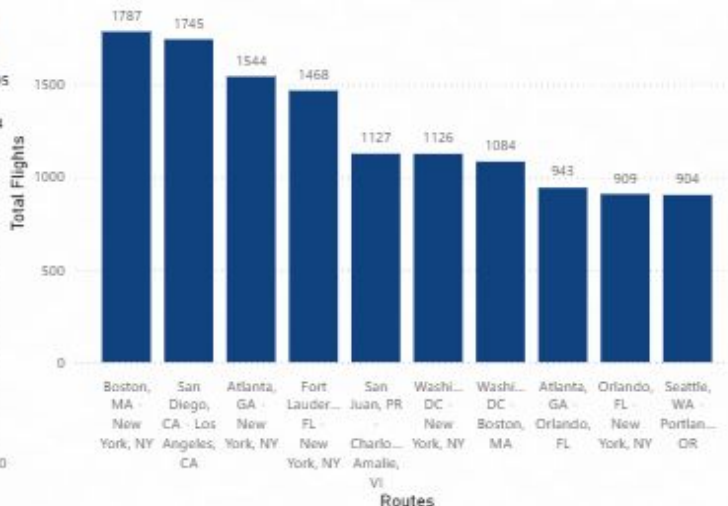
Years

2008	2010	2012
2009	2011	2013

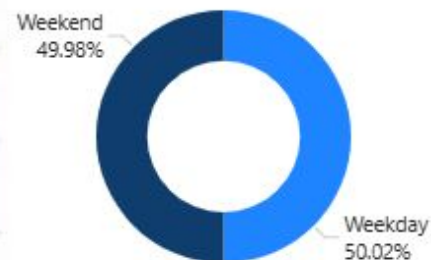
Load Factor % by Carrier Name



Top 10 Performing Routes By Flights



Load Factor % by Day Type



Quarter

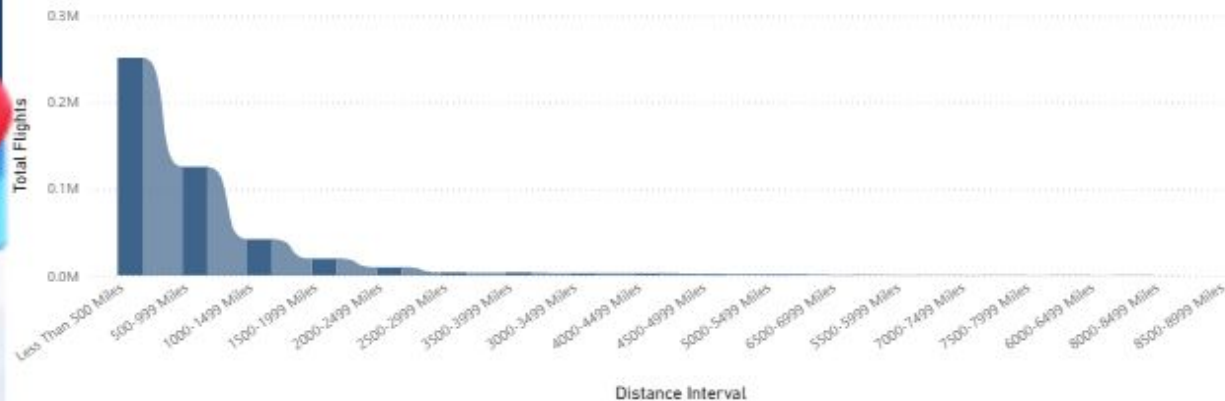
- ☐ Q1
- ☐ Q2
- ☐ Q3
- ☐ Q4

Day Type

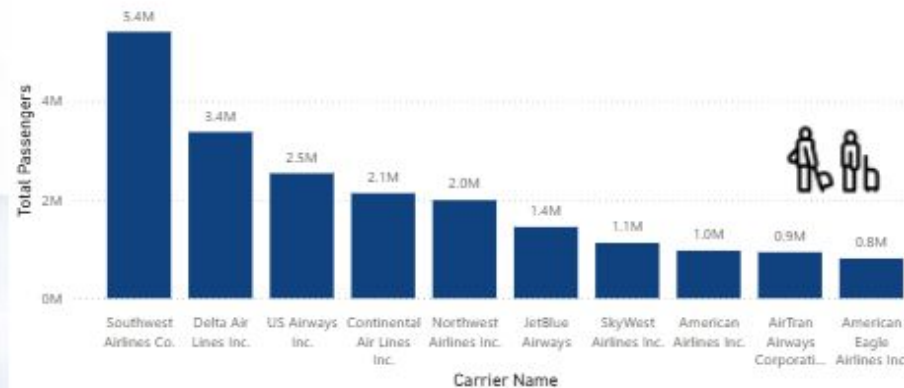
Weekday

Weekend

Distance Based Flight Analysis



Top 10 Carriers by Passenger Performance





Weektype

☒ (All)

☒ Weekday

☒ Weekend

Carrier Name

(All)

Origin Country

(All)

Destination Country

(All)

Origin City

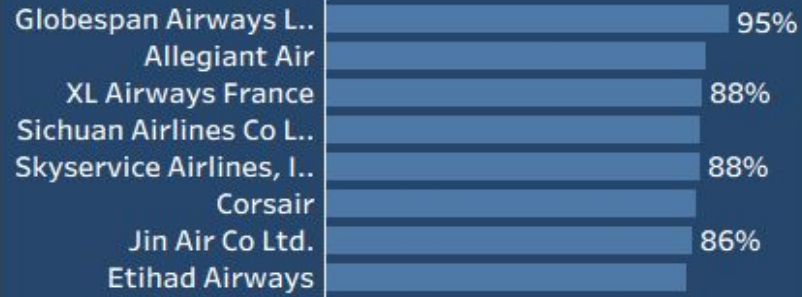
(All)

Destination City

(All)

Carrier wise Load Factor

Carrier Name

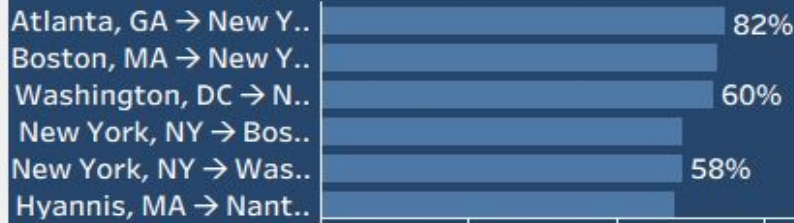


0.0 0.2 0.4 0.6 0.8 1.0

Load Factor

Top Routes

Route field



0K 2K 4K 6K

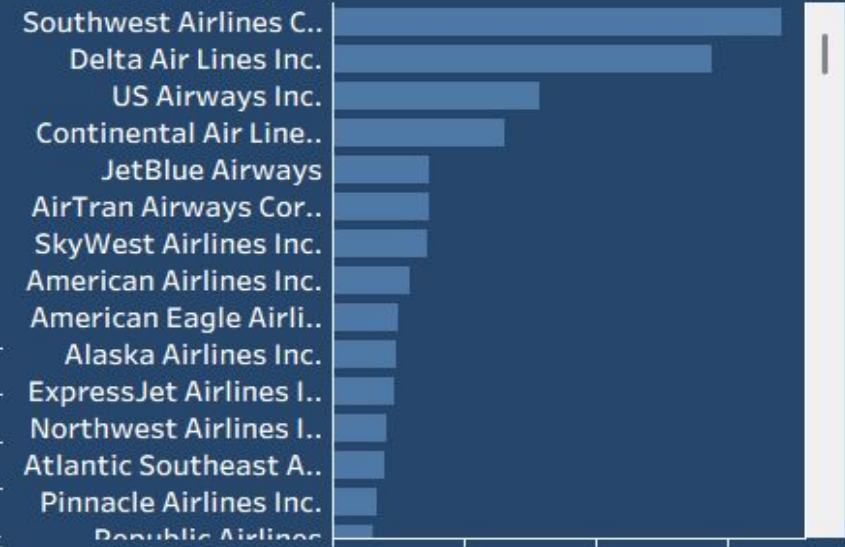
Departures Performed

Weekday vs Weekend Load Factor



Top 10 Carriers

Carrier Name



0M 10M 20M 30M

Transported Passengers

Flights By Distance Group

%Distan..



0K 500K 1000K 1500K 2000K

Departures Performed

High Cloud Airline P..

Load Factor by year





High Cloud Airlines

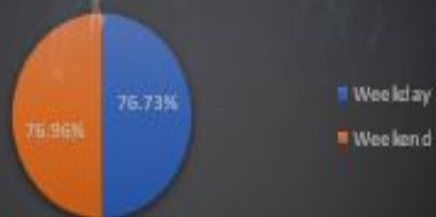
Total transported passenger:	Total flights performed	Load factor %	Total freight transported	Total Distance travelled	Avg passenger per flight	Weekend load factor %	Weekday load factor %
187022609	2799356	76.80%	8492011117	82338802	66.80915503	76.96%	76.73%



Flights on distance group



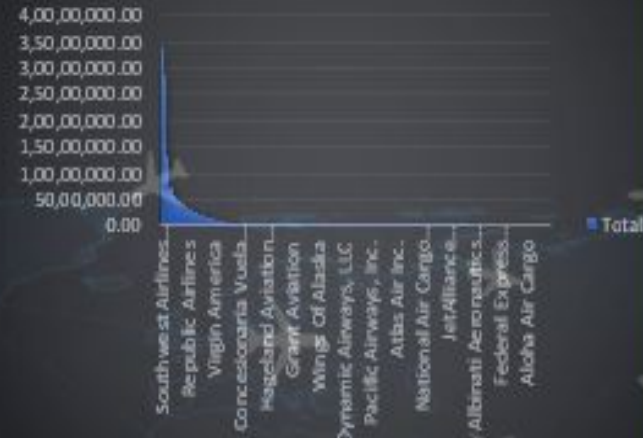
Weekday V/s weekend



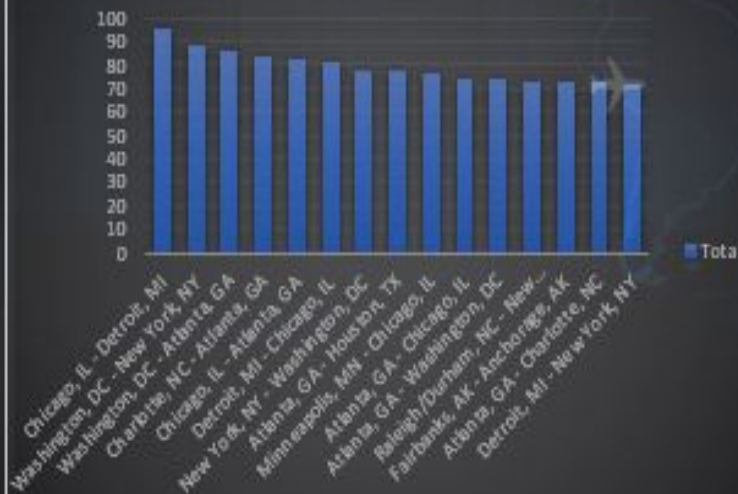
Carrier name



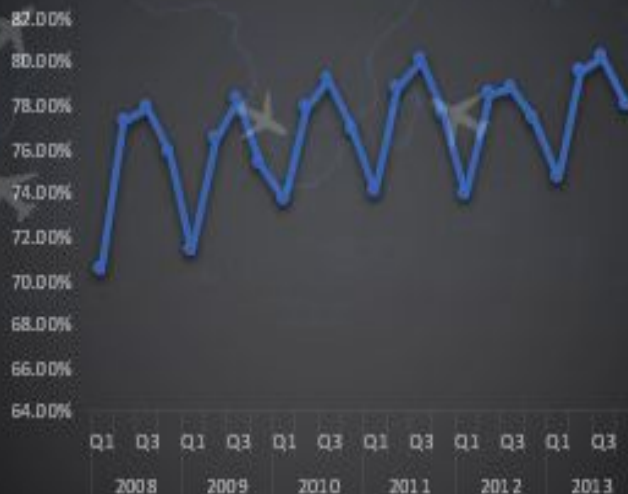
Total passengers



Top routes



Load Factor



Origin City	Destination
47-Mile Mine,...	47-Mile Min...
Abbotsford, C...	Abbotsford, ...
Aberdeen, ID	Aberdeen, SD
Aberdeen, SD	Abilene, TX
Abilene, TX	Abu Dhabi, ...
Origin St...	Destination
(blank)	(blank)
Alabama	Alabama
Alaska	Alaska
Alberta	Alberta
Arizona	Arizona
Arkansas	Arkansas
Origin C...	Destinati...
Angola	Afghanistan
Anguilla	Angola
Antigua and ...	Anguilla
Argentina	Antigua and ...
Aruba	Argentina
Australia	Aruba
Austria	Australia

SQL KPI's

Limit to 1000 rows											
<pre>19 -- Table structure for table `maindata` 20 -- 21 22 • DROP TABLE IF EXISTS `maindata`; 23 • /*!40101 SET @saved_cs_client = @@character_set_client */; 24 • /*!50503 SET character_set_client = utf8mb4 */; 25 • CREATE TABLE `maindata` (26 `%Airline ID` int DEFAULT NULL, 27 `%Carrier Group ID` int DEFAULT NULL, 28 `%Unique Carrier Code` text</pre>											
Result Grid											
Filter Rows: Export: Wrap Cell Content: Fetch rows:											
	%Airline ID	%Carrier Group ID	%Unique Carrier Code	%Unique Carrier Entity Code	%Region Code	%Origin Airport ID	%Origin Airport Sequence ID	%Origin Airport Market ID	%Origin World Area Code	%Destination Airport ID	%Dest Sequer
	20336	5	H6	6984	D	13873	1387301	33873	1	15721	157210
	20331	5	8E	6992	D	15478	1547801	35478	1	12676	126760
	20455	5	GV	6985	D	12853	1285301	32853	1	11336	113360
	20336	5	H6	6984	D	15478	1547801	35478	1	13873	138730
	20336	5	H6	6984	D	10405	1040501	30405	1	10551	105510
	20336	5	H6	6984	D	10551	1055101	30113	1	12851	128510
	20340	5	K5	6966	D	14797	1479701	34797	1	14797	147970
	20336	5	H6	6984	D	14234	1423401	34234	1	10304	103040
	20225	5	KS	6975	D	10245	1024501	30245	1	13714	137140

SQL KPI's

```
147 • select * from maindata;  
148     -- 2. Find the load Factor percentage on a yearly , Quarterly , Monthly basis ( Transported passengers / Available seats)  
149  
150 • select Year,round(`Month`/3) as Qtr,`Month`,(sum(`# Transported Passengers`)/Sum(`# Available Seats`))*100 as factor from maindata  
151 group by Year,Qtr,`Month` order by Year,`Month`;  
152
```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content: 

	Year	Qtr	Month (#)	factor
	2013	1	2	72.5004
	2013	1	3	78.1951
	2013	1	4	76.9900
	2013	2	5	79.7258
	2013	2	6	81.8032
	2013	2	7	81.9951
	2013	3	8	81.0862
	2013	3	9	77.0772
	2013	3	10	79.2663
	2013	4	11	79.0538

maindata 1 Result 2 × Result 3 Result 4 Result 5 Result 6 Result 7

SQL KPI's

```
152
153     -- 3. Find the load Factor percentage on a Carrier Name basis ( Transported passengers / Available seats)
154
155 •   select `Unique Carrier`,(sum(`# Transported Passengers`)/Sum(`# Available Seats`))*100 as factor from maindata
156     group by `Unique Carrier` order by factor Desc;
157
158     4. Identify Top 10 Carrier Names based passengers preference
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

Unique Carrier	factor
US Airways Inc.	80.0118
JetBlue Airways	79.8454
GoJet Airlines, LLC d/b/a United Express	79.5691
AirTran Airways Corporation	79.5369
ExpressJet Airlines Inc.	78.7969
Hainan Airlines Company Limited	78.5959
Virgin America	78.5924
American Airlines Inc.	78.3654
Meridiana S.p.A	78.3425
LAN Argentina	78.3168
Alaska Airlines Inc.	78.3030

SQL KPI's

```
158      -- 4. Identify Top 10 Carrier Names based passengers preference
159
160 •   select `Unique Carrier`,sum(`# Transported Passengers`) as total_passengers from maindata
161      group by `Unique Carrier` order by total_passengers desc limit 10;
```

Result Grid   Filter Rows: | Export:  | Wrap Cell Content:  | Fetch rows: 

	Unique Carrier	total_passengers
▶	Southwest Airlines Co.	34107505
	Delta Air Lines Inc.	28837271
	US Airways Inc.	15646967
	Continental Air Lines Inc.	13123800
	JetBlue Airways	7312002
	AirTran Airways Corporation	7299774
	SkyWest Airlines Inc.	7126582
	American Airlines Inc.	5900880
	American Eagle Airlines Inc.	4957403
	Alaska Airlines Inc.	4883403

SQL KPI's

```
162
163 -- 5. Display top Routes ( from-to City) based on Number of Flights
164
165 • select concat(`Origin City`,`-`,`Destination City`) as Route,count(`%Airline ID`) as Number_of_flights from maindata
166 group by Route order by Number_of_flights desc;
167
168 7. Identify number of flights based on Distance group
```

Result Grid   Filter Rows: Export:  Wrap Cell Content:  Fetch rows: 

	Route	Number_of_flights
►	Chicago, IL-Detroit, MI	95
	Washington, DC-New York, NY	88
	Washington, DC-Atlanta, GA	86
	Charlotte, NC-Atlanta, GA	83
	Chicago, IL-Atlanta, GA	82
	Detroit, MI-Chicago, IL	81
	New York, NY-Washington, DC	78
	Atlanta, GA-Houston, TX	78
	Minneapolis, MN-Chicago, IL	77
	Atlanta, GA-Chicago, IL	74
	Atlanta, GA-Washington, DC	74

SQL KPI's

```
173      -- 6. Identify the how much load factor is occupied on Weekend vs Weekdays.
174
175 • Select CASE WHEN DAYOFWEEK(STR_TO_DATE(CONCAT(Year, '-', `Month` (#), '-', Day), '%Y-%m-%d')) IN (1,7)
176 THEN 'Weekend' ELSE 'Weekday' END AS day_type, (SUM(`# Transported Passengers`) / SUM(`# Available Seats`))*100 AS load_factor
177 FROM maindata GROUP BY day_type;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	distance_group	number_of_flights
▶	1	58047
	2	28131
	3	11333
	4	6092
	5	2330
	8	1084
	6	855
	9	810
	7	612
	10	524
	11	419

maindata 1 | Result 2 | Result 3 | Result 4 | Result 5 | Result 6 x | Result 7

SQL KPI's

```
168 -- 7. Identify number of flights based on Distance group
169
170 • select distinct(`%Distance Group ID`) as distance_group,count(`%Airline ID`) as number_of_flights from maindata
171 group by `%Distance Group ID` order by number_of_flights desc;
172
173 -- 6. Identify the how much load factor is occupied on Weekend vs Weekdays.
174
```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	day_type	load_factor
▶	Weekend	76.9557
	Weekday	76.7306

maindata 1 Result 2 Result 3 Result 4 Result 5 Result 6 Result 7 ×

Project Conclusion

The implementation of a modern BI stack has transformed how we approach strategic decision-making.

- **Strategic Performance Insights:** Evaluated 2008–2013 data to uncover trends in load factor, revenue, delays, and route performance.
- **Operations Optimization:** Diagnosed key operational bottlenecks and optimized routes and scheduling for profitability.
- **Actionable Visualizations:** Developed interactive dashboards and robust SQL-driven KPIs to guide real-time decision-making.
- **Experience & Efficiency Gain:** Provided concrete methods to enhance passenger experience while driving overall airline efficiency.



"Data is only as valuable as the decisions it enables."

Thank you for your time. The comprehensive dashboards and SQL documentation have been attached in the project appendix for your review.